

MATH 899 - INDP Categories & Sheaves

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0.1 Categories & Functors

0.1.1 What is a category

Definition 0.1 (Category). *A category contains 4 main things.*

- *A collection of data composed of sets*
- *A collection of morphisms between sets in data*
- *The set of morphisms must contain some kind of identity function such that each set in the data collection could be mapped to itself.*
- *There must be some way to compose morphisms subject to the following axioms:*

– **Associativity** : *For any composite morphism such as $F \xrightarrow{f} G \xrightarrow{g} B \xrightarrow{h} C$ then*

$$h \circ (f \circ g) = (h \circ g) \circ f$$

– **Identity** : *For any morphism f , any composite formed by composing the morphism with the identity should be equal to f .*

Interpretation : What I understand a category as is a way to generalize the idea of a set. Sets are usually defined by objects within the set and certain morphisms (usually functions) that relate all or nearly all objects in the set to each other. A basic example would be a group. Under the category definition we could redefine it as a category where the data is simply one set, the elements contained in the group, and the morphisms as the group operations. By definition those morphisms include their identities because those morphisms are isomorphisms and so follow the rules of function composition. What seems to be more powerful is the ability to take more than one set and put them definitively categorize them with some overarching behavior.

0.1.2 Examples of Categories

Set : The most basic example of categories is **Set**. In this category the data is a collection sets and the morphisms are functions between sets.

Pre/Pos Consider two preorders $(X, \leq_X), (Y, \leq_Y)$. We could generate the category of preorder **Pre** where our data is a collection of preorders and our morphisms are monotone order-preserving maps between any two preorders. In the same way we can generate the category of posets **Pos** where the data is a collection of posets and our morphisms are still monotone order-preserving maps.

Grp The category of groups consists of groups where morphisms are group homomorphisms. **Note:** A group G always has a binary operation $*$: $G \times G \rightarrow G$. A binary operation, under a group, carries the associativity and identity properties required to be a morphism in a category. With this idea we can define a category **C** where objects are elements of G and morphisms between elements are given by the binary operation. Because of the nature of the binary operation under G we can create a well defined category. This leads to the notion of a small category, a category whos set of objects contains individual elements rather than sets.

Vect_k The category of vector spaces is a category consisting of a vector spaces and whose morphisms are the set of linear transformations between vector spaces. Like a group, a finite dimensional vector space over k can define a small category where morphisms are linear transformations of vectors.

Top The category of topological spaces is a category consisting of topological spaces where morphisms are continuous maps between spaces.

Mod_R The category of R -Modules is a category modules over a ring R where morphisms are module homomorphisms.

Ring The category of rings is a category consisting of rings where morphisms are ring homomorphisms.

0.1.3 What is a functor

Given that I can now categorize things both literally and mathematically, how do I start creating relationships between categories? The power of categories comes from functors which act as a generalization between functions, mapping between two categories in a manner similar to that a function mapping between two sets.

Definition 0.2 (Covariant Functor). *A functor $F : C \rightarrow D$ between categories C, D is an assignment of*

- (i) *an object $F(c) \in D$ for every $c \in \text{Obj}(C)$*
- (ii) *a morphism $F(f) : F(C) \rightarrow F(C')$ in D for every morphism $c \rightarrow c'$ in C*

satisfying the axioms

(a) For any object $c \in C$,

$$F(id_c) = id_{F(c)}$$

(b) For any composable pair f, g in C

$$F(g) \circ F(f) = F(g \circ f)$$

Represented by the commutative diagrams

$$\begin{array}{ccc} c & \xrightarrow{f} & c' \\ & \searrow g \circ f & \downarrow g \\ & & c'' \end{array} \quad \text{inducing} \quad \begin{array}{ccc} F(c) & \xrightarrow{F(f)} & F(c') \\ & \searrow F(g \circ f) & \downarrow F(g) \\ & & F(c'') \end{array}$$

Definition 0.3 (Contravariant Functor). A contravariant is a functor F between categories subject to the same axioms BUT the direction of the target morphisms are flipped.

$$\begin{array}{ccc} id_A \hookrightarrow A & & F(A) \\ \downarrow f(A) & \text{red } F(f) \downarrow & \\ (g \circ f)(A) \curvearrowright B & & F(B) \\ \downarrow g(B) & \text{red } F(g) \downarrow & \\ C & & F(C) \end{array} \quad \text{red } F(g \circ f) \quad \begin{array}{ccc} id_A \hookrightarrow A & & F(A) \\ \downarrow f(A) & \text{red } F(f) \uparrow & \\ (g \circ f)(A) \curvearrowright B & & F(B) \\ \downarrow g(B) & \text{red } F(g) \uparrow & \\ C & & F(C) \end{array} \quad \text{red } F(g \circ f)$$

$$M \xrightarrow{Cov} N$$

$$M \xrightarrow{Contra} N$$

In some cases the direction of the contravariant is preferred because we are preserving the arrow directions of the source category (see natural transformation dual vectors example). An important example is the covariant hom-functor $\text{Hom}_C(A, -) : \mathbf{C} \rightarrow \mathbf{Set}$. Consider the following morphism diagram where $A, X, Y \in \text{Obj}(C)$

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ & \searrow g \circ f & \downarrow g \\ & & Y \end{array}$$

In each individual case you can apply composition of function to move from $X \rightarrow A$. How can we approach this in a more general manner? Rephrasing this question, how can we represent the process of function composition on any arbitrary set. Consider the case where $\mathbf{C} = \mathbf{Set}$ and then our functor takes the form $\text{Hom}_{\mathbf{Set}}(A, -) : \mathbf{Set} \rightarrow \mathbf{Set}$. Fix some $x \in \text{Obj}(C)$, then $\text{Hom}_{\mathbf{Set}}(A, X)$

returns the set of morphisms between X and A . If you fix a morphism $f : X \rightarrow Y$ in $\mathbf{C}$ then when applying the functor

$$\text{Hom}_{\mathbf{Set}}(A, f) : (A \rightarrow X) \rightarrow (A \rightarrow Y)$$

So when passing in f we get out $g \circ f$. The general functor can be illustrated with the commutative diagram:

$$\begin{array}{ccc} X & \xrightarrow{\text{Hom}_{\mathbf{C}}(A, -)(X)} & \text{Hom}_{\mathbf{C}}(A, X) \\ f \downarrow & & \downarrow \text{Hom}_{\mathbf{C}}(A, -)(f) \\ Y & \xrightarrow{\text{Hom}_{\mathbf{C}}(A, -)(Y)} & \text{Hom}_{\mathbf{C}}(A, Y) \end{array}$$

Notice that the direction of the arrows is being preserved and precisely because we are interested in preserving the direction of composition. If, for example, we are interested in representing precomposition then we would like the resulting arrows after the application of our functor being reversed. This is defined as the contravariant functor $\text{Hom}_{\mathbf{C}}(-, A) : \mathbf{C}^{op} \rightarrow \mathbf{Set}$.

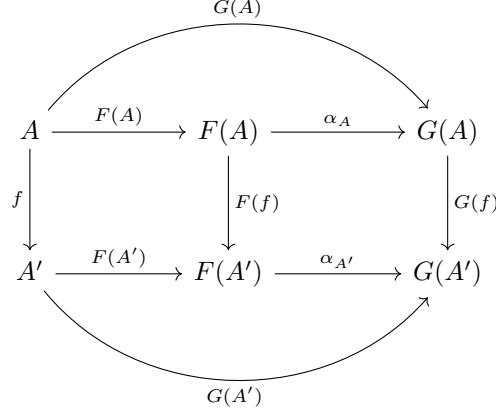
0.2 Natural Transformations of Functors

Now we have the notion of using functors to make comparisons between different categories. But what happens when we can define multiple different functors to make the same comparison. I think you would logically ask yourself whether there is some kind of shared property between the functors that would allow us to compare them. This is where the notion of a natural transformation becomes useful. Consider two functors $F, G : \mathbf{C} \rightarrow \mathbf{D}$ how can we encode the relationship between F and D .

Definition 0.4 (Natural Transformation). *Given two categories $\mathbf{C}, \mathbf{D}$ and functors $F, G : \mathbf{C} \rightarrow \mathbf{D}$ a natural transformation $\alpha : F \Rightarrow G$ is a collection of morphisms called $\mathbf{C}$ -components denoted $\alpha_c : F(C) \rightarrow G(C)$ indexed by all objects in $\mathbf{C}$. Diagrammatically:*

$$\begin{array}{ccc} & F & \\ \curvearrowright & & \searrow \\ C & \Downarrow \alpha & D \\ \curvearrowleft & & \nearrow \\ & G & \end{array}$$

An clearer diagram for a given $A, A' \in \mathbf{C}$



Note that the $\mathbf{C}$ -components live within $\mathbf{D}$. Now we aim to apply generalizations of function processes to natural transformations

Definition 0.5 (Natural Isomorphism). *A natural isomorphism is a natural transformation $\alpha : F \Rightarrow G$ such that each component has an inverse. This is denoted $\alpha : F \cong G$*

Now equipped with the notion of natural transformations and natural isomorphisms we can apply them to compare categories and more importantly determine their equivalence.

Definition 0.6 (Equivalence of Categories). *Consider two categories $\mathbf{C}, \mathbf{D}$ with two functors $F : \mathbf{C} \rightarrow \mathbf{D}$ and $G : \mathbf{D} \rightarrow \mathbf{C}$ with natural isomorphisms $\eta : id_{\mathbf{C}} \cong G \circ F$ and $\epsilon : id_{\mathbf{D}} \cong F \circ G$. The categories are then considered to be equivalent if there exists an equivalence of categories between them. This denoted $\mathbf{C} \simeq \mathbf{D}$.*

At this point, nothing is stopping us from considering a category of functors. Where objects are functors and morphisms are natural transformations between those functors. For any fixed categories $\mathbf{C}, \mathbf{D}$ we can form the functor category $\mathbf{C}^{\mathbf{D}}$ as the category of functors from $\mathbf{C}$ to $\mathbf{D}$ where the morphisms are natural transformations between those functors.

0.3 Limits & Colimits

Definition 0.7 (Limit). *Let $\mathbf{J}$ be an indexing diagram of category $\mathbf{C}$ and the functor $F : \mathbf{J} \rightarrow \mathbf{C}$. The limit of the diagram is an object $\lim_{\mathbf{J}} A$ of $\mathbf{C}$ equipped with morphisms $f_j : \lim_{\mathbf{J}} A \rightarrow A_j$ for each $j \in \mathbf{J}$ such that if $m : j \rightarrow k$ is a morphism in $\mathbf{J}$, then*

$$\begin{array}{ccc} \lim_{\mathbf{J}} A & & \\ f_j \downarrow & \searrow f_k & \\ A_j & \xrightarrow{F(m)} & A_k \end{array}$$

commutes. If this limit exists then it is unique up to isomorphism.

Definition 0.8 (Colimit). *By the symmetric nature of categories if $\mathbf{I}$ is the indexing category of $\mathbf{C}$. Then there exists a functor $F : \mathbf{I}^{op} \rightarrow \mathbf{C}$. Then there exists an object $\text{colim}_{\mathbf{J}} A$ with morphisms $f_j : A_j \rightarrow \text{colim}_{\mathbf{J}} A$ for each $j \in \mathbf{J}$ such that if $m : j \rightarrow k$ is a morphism in $\mathbf{J}$ then*

$$\begin{array}{ccc} A_j & \xrightarrow{F(m)} & A_k \\ f_j \uparrow & \nearrow f_k & \\ \text{colim}_{\mathbf{J}} A & & \end{array}$$

commutes. If this colimit exists then it is unique up to isomorphism.

0.3.1 Examples of Limits

Poset Suppose that a partially ordered set $(\mathcal{I}, \leq)$ has an initial object e . Then for there exists some category $\mathbf{C}$ where $F : \mathbf{J} \rightarrow \mathbf{C}$ is the realization of this diagram. Then there exists an object $\lim_{\mathbf{J}} F = F(e)$ there exists, under the poset relation, a map $\leq_j : e \rightarrow F(j)$ for each $j \in \mathbf{J}$. The under the poset relation there exists $\leq_m : j \rightarrow k$ in $\mathbf{J}$ such that

$$\begin{array}{ccc} F(e) & & \\ \leq_j \downarrow & \searrow \leq_k & \\ F(j) & \xrightarrow{F(\leq_m)} & F(k) \end{array}$$

which commutes under the transitive properties of a poset relation.

Set Let $\mathcal{I}$ be the indexing category of **Set**. The object

$$\left\{ (a_i)_{i \in \mathcal{I}} \in \prod_i A_i : F(m)(a_j) = a_k \text{ for all } m \in \text{Mor}_{\mathcal{I}}(i, j) \in \text{Mor}(\mathcal{I}) \right\}$$

denoted $\lim_{\mathcal{I}} A$ is the limit of **Set**. This object states that taking the cartesian product of all sets such that there exists an $m : A_j \rightarrow A_k$ taken to a corresponding $F(m)$. This object equipped with the projection maps $\pi_j : \lim_{\mathbf{J}} A \rightarrow A_j$ for all $j \in \mathcal{I}$. Then for each $m : j \rightarrow k$ the following diagram

$$\begin{array}{ccc} \lim_{\mathbf{J}} A & & \\ \pi_j \downarrow & \searrow \pi_k & \\ A_j & \xrightarrow{F(m)} & A_k \end{array}$$

clearly commutes $(\pi_j \circ F(m))(a) = a_k = \pi_k(a)$ for all $a \in \lim_{\mathbf{J}} A$.

Top Let $\mathcal{I}$ be the indexing category of **Top** such that $F : \mathcal{I} \rightarrow \mathbf{Top}$ is the realization of the diagram. The $\lim_{\mathcal{I}} A$ is the cartesian product $\prod_{j \in \mathcal{I}} X_j$ equipped with the product topology. Much like the cartesian product of a set, the mappings from the limit to each object will be the canonical projection maps $p_i : \prod_{j \in \mathcal{I}} X_j \rightarrow X_i$ described by the product topology.

0.4 Sheaves

0.4.1 Presheaves

Definition 0.9 (Presheaf). Let $\mathcal{O}(X)$ denote the category of open subsets of a topological space X . A presheaf is a functor $P : \mathcal{O}(X)^{op} \rightarrow \mathbf{Set}$. For each inclusion of open sets $U \hookrightarrow V$ there is a restriction map

$$res_{U,V} : P(V) \rightarrow P(U)$$

that satisfies the following properties

- (i) $res_{U,U} = id_{P(U)}$
- (ii) If $U \hookrightarrow V \hookrightarrow W$ the following diagram commutes

$$\begin{array}{ccc} P(W) & \xrightarrow{res_{W,V}} & P(V) \\ & \searrow res_{W,U} \quad \swarrow res_{V,U} & \\ & P(U) & \end{array}$$

A simple example would be the set of continuous real-valued functions on $\mathbb{R}$, denoted $C(\mathbb{R})$. Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$. Now consider the functor

$$P : \mathcal{O}(C(\mathbb{R}))^{op} \rightarrow \mathbf{Set}$$

it's clear that for any inclusion of open sets $U \hookrightarrow V$ we can define a restriction function that satisfies the presheaf conditions. As we will see in the next section the definition of a sheaf formalizes the relationship between local and global information. Part of this story is the need to understand behavior at every point of a space in order to properly link the behavior to a global picture. This naturally leads us to the definition of a stalk.

Definition 0.10 (Stalk of a Presheaf). If $\mathcal{F}$ is a presheaf on X . The stalk $\mathcal{F}_p$ of $\mathcal{F}$ is defined as the colimit over all open sets $U \subseteq X$ containing p

$$\mathcal{F}_p = \text{colim}_{p \in U} \mathcal{F}(U)$$

Ideally we would want these local stalks to define the entire presheaf, but we don't have enough structure yet. We need an important condition called glueing which we will define when presenting sheaves in the next section. First we will talk about the sections of a presheaf and how they may be interpreted as functions. This will ease us into the notation presented in the definition of a sheaf.

Definition 0.11 (Section). Let $\mathcal{F}$ be a presheaf on some topological space X . A section is an element $s|_U$ of $\mathcal{F}(U)$ for some open subset of X .

As we have defined above the stalk of a sheaf $\mathcal{F}_x$ of a sheaf $\mathcal{F}$ is a colimit taken over all open neighborhoods of V of a point x . This is by definition along the restriction $\rho : U \rightarrow U'$. More concretely, we can express the stalk as

$$\mathcal{F}_x = (\bigcup_{x \in U} F(U) / \sim)$$

where two sections $f \in \mathcal{F}(U)$ and $g \in \mathcal{F}(U')$ are related if there exists and $x \in U'' \subseteq U \cap U'$ and f and g agree when restricted to U'' . The equivalence class in the stalk $\mathcal{F}_x$ is denoted f_x , called the germ of f at x . Since $\mathcal{F}_x$ is the colimit of $\mathcal{F}$ we can consider the function $\alpha : \mathcal{F}(U) \rightarrow \mathcal{F}_x$ where $\alpha(s) \mapsto f_x$. Through this we can interpret a section as associating a point $x \in U$ to a particular germ f_x . More explicitly as a function $s : U \rightarrow \mathcal{F}_x$ where $s(x) \mapsto f_x$.

$$\begin{array}{ccccc}
 x \in & & U & \xhookrightarrow{\subseteq} & V \\
 \vdots & & \downarrow \mathcal{F} & & \downarrow \mathcal{F} \\
 s|_U \in & & \mathcal{F}(U) & \xrightarrow{\rho_U^V} & \mathcal{F}(V) \\
 \vdots & & \downarrow \alpha & \swarrow \alpha' & \\
 f_x \in & & \mathcal{F}_x & &
 \end{array}$$

We have established the mechanism to investigate local behavior of a presheaf. A natural follow up questions is "What happens to the stalks of a presheaf under a morphism".

Definition 0.12 (Morphisms of Presheaves). *Let $\mathcal{F}, \mathcal{G}$ be presheaves and a morphism $\phi : \mathcal{F} \rightarrow \mathcal{G}$ consists of morphisms $\phi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ such that under the inclusion $V \subseteq U$ the diagram*

$$\begin{array}{ccc}
 \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\
 \downarrow \rho_U^V & & \downarrow \rho_V^U \\
 \mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{G}(V)
 \end{array}$$

commutes.

In fact a morphism of presheaves induces a morphism on stalks. Consider the following commutative diagram where ϕ is a morphism of presheaves $\mathcal{F}, \mathcal{G}$.

$$\mathcal{F}_x \longleftarrow \alpha \longrightarrow \mathcal{F}(U) \xrightarrow{\phi(U)} \mathcal{G}(U) \xrightarrow{\beta} \mathcal{G}_x$$

The commutativity of maps to colimits allows us to construct a morphism $\phi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ where $\phi_x = \beta \circ \phi(U) \circ \alpha^{-1}$. The fact that individual morphisms on

stalks doesn't induce a morphism on presheaves is an indication of the local nature of presheaves. Note that in categorical language what we are really investigating is a particular natural transformation $\varphi : \mathcal{F} \Rightarrow \mathcal{G}$.

Like any morphism we aim to capture as much of it's behavior as possible. We do this by studying the kernel, cokernel, and image of φ . Remember that under the hood a morphism between sheaves is really a natural transformation between sheaves. This means that φ is really representing a family of morphisms between objects in the image of $\mathcal{F}$ and $\mathcal{G}$. We can use this to define the following:

Definition 0.13 (Kernel, Cokernel, Image Presheaves). *The kernel presheaf is given by $U \mapsto \ker \varphi(U)$. The image presheaf is given by $U \mapsto \text{im } \varphi(U)$. The cokernel presheaf is given by $U \mapsto \text{coker } \varphi(U)$.*

0.4.2 Sheaves

With the notion of sections, we may now begin to impose more structural conditions on presheaves in an attempt to preserve more information about the relationships in the topological space we started with.

Definition 0.14 (Sheaf). *Let $\mathcal{O}(X)$ be the category of open subsets of a topological space X , where objects are open subsets of X and morphisms are inclusions of one subset into another. Given a presheaf $F : \mathcal{O}(X)^{op} \rightarrow \mathbf{Set}$, an open set U with an open cover by $\{U_i\}_{i \in I}$, indexed by a I , where $s_i \in F(U_i)$, then F is a sheaf provided it satisfies both:*

1. (Existence/Gluing) *If for each i , there is a section $s_i \in F(U_i)$ satisfying that for each pair U_i and U_j the restriction of s_i and s_j to the overlap $U_i \cap U_j$*

$$s_i(x) = s_j(x)$$

for all $x \in U_i \cap U_j$ for all i, j . Then, there exists a section $s \in F(U)$ with restrictions $s|_{U_i} = s_i$ for all i .

2. (Uniqueness/Locality) *If $s, t \in F(U)$ is such that*

$$s|_{U_i} = t|_{U_i}$$

for all i , then $s = t$.

Interpreting definition (1.12) in terms of sheaves, we can treat a morphism of sheaves exactly the same way we did with presheaves. The kernel, cokernel, and image are all still presheaves, but the kernel is always a sheaf, and, generally, cokernels and images are not sheaves. To check that the kernel is a sheaf we check the following conditions:

Gluing Consider an open subset U of a topological space X . This open set has a cover $\{U_i\}_{i \in I}$ such that each $x_i \in \ker(\varphi(U_i))$ agree across intersections. We can interpret each these x_i as local sections of $\mathcal{F}(U_i)$. Since $\mathcal{F}$ is a sheaf, we can glue each of these local sections to build a section $x \in \mathcal{F}(U)$.

To finish this portion it suffices to show that the section $x \in \ker(\varphi(U))$. Note that φ is a morphism of sheaves so the inclusion $U_i \subseteq U$ generates the diagram

$$\varphi_U(x) \in \mathcal{F}(U) \xrightarrow{\rho|_{U_i}^{U_i}} \mathcal{F}(U_i) \ni \varphi_U(x)|_{U_i} = \varphi_{U_i}(x_i)$$

i.e. the morphism of sheaves commutes under the restriction of sections. Since each $\varphi_{U_i}(x_i) = 0$ in $\mathcal{G}(U_i)$, then the sheaf condition on $\mathcal{G}$ guarantees that the global section $x \in \ker(\varphi(U))$.

Uniqueness Since $x \in \ker(\varphi(U))$ is an element of $\mathcal{F}(U)$, then uniqueness follows from the fact that $\mathcal{F}$ is a sheaf.

Recall that a morphism of presheaves induces a morphism between stalks, but the local nature of a presheaf prevent a morphism of stalks from inducing a morphism on presheaves. The global nature of sheaves removes this obstruction.

Theorem 0.1. *Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves. Then φ is an isomorphism of sheaves if and only if the induced map on the stalk $\varphi_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$ is an isomorphism for every $p \in X$.*

Proof. If φ is an isomorphism it is clear that each morphism of stalks of φ_p must be an isomorphism. Now suppose that φ_p is an isomorphism. Suppose that for a section s of U that $\varphi(s)$ is 0 in $\mathcal{G}$. Consider that for every point $p \in U$ that the assumption that $\varphi(s)$ is 0 implies that the image of each stalk $\varphi(s)_p$ in $\mathcal{G}_p$ is 0. Since φ_p is injective, it follows that s_p is 0 in $\mathcal{F}_p$. By the definition of a stalk this means that there exists an open neighborhood of p , $W_p \subseteq U$ such that the restriction of $s|_{W_p}$ is 0 in $\mathcal{F}(U)$. Since U can be covered by W_p for all points $p \in U$, the gluing property of sheaves implies that s must be 0 in $\mathcal{F}(U)$. Therefore φ is injective. Towards surjectivity, suppose that we have a section $s \in \mathcal{G}(U)$ with a germ $s_p \in \mathcal{G}_p(U)$ at a point p . Since φ_p is surjective, there exists germ $t_p \in \mathcal{F}_p(U)$ such that $\varphi(t_p) = s_p$. s_p can be represented by some section S_p on an open neighborhood V_p of a point p . Then $\varphi(S_p)$ and $t|_{V_p}$ are both elements of $\mathcal{G}(V_p)$ whose germs at a point p must be identical. Therefore we can assume that $\varphi(S_p) = t|_{V_p}$. We know that U can be entirely covered by V_p for every point $p \in U$. Now pick two points $p, q \in U$ with open neighborhoods $V_p, V_q \subseteq U$. Then the sections $S_p|_{V_p \cap V_q}$ and $S_q|_{V_p \cap V_q}$ are sections of $\mathcal{F}(V_p \cap V_q)$ which are both sent to $t|_{V_p \cap V_q}$ by φ . Following from the injectivity we have proved above $S_p = S_q$. Then by the gluing property there exists a section $s \in \mathcal{F}(U)$ such that $s|_{V_p} = S_p$ for each p . Since $\varphi(s)$ and t are both sections of $\mathcal{G}(U)$. For each $p \in U$ the sections $\varphi(s)|_{V_p} = t|_{V_p}$. By the sheaf uniqueness property we can conclude that $\varphi(s) = t$. $\square$

Following this result we can talk about one of the most important properties of sheaves. In particular this following theorem can be interpreted as a universal property of presheaves in a categorical sense.

Theorem 0.2. *Given a presheaf $\mathcal{F}$ there exists a sheaf $\mathcal{F}^+$ and a morphism $\theta : \mathcal{F} \rightarrow \mathcal{F}^+$, with the property that for any sheaf $\mathcal{G}$ and morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, there is a unique morphism $\psi : \mathcal{F}^+ \rightarrow \mathcal{G}$ such that $\varphi = \psi \circ \theta$. Both $\mathcal{F}^+$ and θ are unique up to unique isomorphism. $\mathcal{F}^+$ is called the sheaf associated to $\mathcal{F}$.*

We see that sheaves are capable of capturing local information and then gluing it into a global picture. But using sets as our global picture leaves us with little room to maneuver after defining the sheaf. So far we have been using entirely algebraic language to describe our sheaves limiting the possible applications. What we will do now is redefine sheaves entirely in terms of categorical language allowing us to broaden our "sheaf" horizon.

To begin, let's analyze the category $\mathcal{O}(X)$. For any open set U with a family of I -indexed open subsets covering U , note that

$$U_i \hookleftarrow U_i \cap U_j \hookrightarrow U_j$$

We can see that the limit(colimit in $\mathcal{O}(X)^{op}$) will be U . Since our sheaf functor is contravariant, colimits are sent to limits in **Set**. This can be expressed with the equalizer diagram:

$$F(U) \xrightarrow{\rho|_{U_i}^U} \prod_{i \in I} F(U_i) \xrightleftharpoons[p]{p} \prod_{i,j \in I} F(U_i \cap U_j)$$

where p is $\rho|_{U_i \cap U_j}^{U_i}$ composed with a projection map, and q is $\rho|_{U_i \cap U_j}^{U_j}$ composed with a projection map. Now we can define a purely categorical sheaf

Definition 0.15 (Sheaf (Categorically)). *Let X be a topological space, and let $\mathbf{D}$ be a category with limits. Let $F : \mathcal{O}(X)^{op} \rightarrow \mathbf{D}$ be a presheaf. If any open set U with an I -indexed family of sets covering U has U as the colimit and the presheaf sends the colimit to a limit in **Set**, then F is a sheaf.*

With this definition, we are now equipped with a local-to-global passage that takes us, not only, to the sparse world of sets but also to richer and more expressive categories of algebraic objects. We will see examples of sheaves defined "categorically" in the section on sheaf cohomology.

Now consider the case where we have sheaves over two different topological spaces X, Y with some continuous map $X \rightarrow Y$. The fact that each open subset of Y has a pre-image in X should intuitively relate to some relationships between the sheaves over these topological spaces. This relationship is called the pushforward of a sheaf (CITE proof under Textbook Exercises # 2).

Definition 0.16 (Pushforward of a Sheaf). *Let $\pi : X \rightarrow Y$ be a continuous map, and F be a presheaf on X . Then define a presheaf $\pi_* F$ on Y such that for any open subset U , set $\pi_* F(U) = F(\pi^{-1}(U))$. If F is a sheaf, then $\pi_* F$ is also a sheaf.*

The pushforward can be imagined as defining a sheaf by pushing it forward through the continuous map π . One rather simple example is provided in textbook examples # 1, called the skyscraper sheaf. A logical followup question is

whether we can define a sheaf on Y and pull our sheaf back through π to X . This can be handled with the pullback of a sheaf.

Definition 0.17 (Pullback of a Sheaf). *Let $\pi : X \rightarrow Y$ be a continuous map, and F be a presheaf on Y . Then define a presheaf π^*F on X such that for any open subset U , set $\pi^*F(U) = F(\pi(U))$. If F is a sheaf, then π^*F is also a sheaf.*

0.4.3 Examples of Sheaves

Continuous Functions Let X be the space of continuous functions on $\mathbb{R}$. An open subset $U \subseteq X$ is defined $U := \{f : U \rightarrow \mathbb{R} | f \text{ is continuous}\}$. Let $F : \mathcal{O}(X)^{op} \rightarrow \mathbf{Set}$ be a presheaf. Given $U \subseteq V \subseteq V$ define the inclusion function $i : U \hookrightarrow V$. Under our presheaf we get the following commutative square.

$$\begin{array}{ccc} U & \xrightarrow{F} & F(U) := \{f : U \rightarrow \mathbb{R} | f \text{ is continuous}\} \\ \uparrow i & & \uparrow F(i) \\ V & \xrightarrow{F} & F(V) := \{g : V \rightarrow \mathbb{R} | g \text{ is continuous}\} \end{array}$$

Then for any $s \in F(U)$, $F(i)(s) = s|_V$ where $s|_V$ is in fact the restriction of a function on V to a function on U .

Manifolds A manifold can be thought of as a "curved" space that locally, in the neighborhood of some point, looks like $\mathbb{R}^n$. That is that a point has an open neighborhood U that is homeomorphic under ϕ to a subset of $\mathbb{R}^n$. The collection $\{(U_i, \phi_i) | i \in I\}$ forms a chart and the collection of charts is called an atlas. A manifold M is covered by a collection of atlases. Given $U_i, U_j \subset M$ where $U_i \cap U_j \neq \emptyset$, it is possible to define a change of coordinates function which preserves the intersection between U_i and U_j in the manifold. In particular a manifold supports a special structure sheaf of smooth function on $\mathbb{R}$ -algebras.

Bundles to (Pre)Sheaves

Definition 0.18 (Bundle, Section). *A bundle over X , X is known as the base space, is a topological space Y equipped with the continuous function $p : Y \rightarrow X$. A section of a bundle p is a function $s : U \rightarrow Y$ such that $p \circ s = id_U$.*

Suppose that you have a bundle over X and an open subset $U \subseteq X$. There exists a restriction of the section map to U , giving us a diagram

$$\begin{array}{ccc} p^{-1}(U) & \xrightarrow{\quad} & Y \\ \downarrow p & \nearrow s & \downarrow p \\ U & \xrightarrow{i} & X \end{array}$$

We can define a collection of sections of our bundle

$$\Gamma_p(U) := \{s \mid s : U \rightarrow Y, p \circ s = id_U, U \subseteq X\}$$

You may notice right away that for an inclusion $U \subseteq V$ we automatically get an inclusion from $\Gamma_p(U) \hookrightarrow \Gamma_p(V)$. In fact we can actually define a sheaf

$$\Gamma_p(-) : \mathcal{O}^{op}(X) \rightarrow \mathbf{Set}$$

Since any two bundles Y, Y' over X are topological spaces, it is possible to define some function $f : Y \rightarrow Y'$. Then we can build the category of bundles over X , $\mathbf{Bn}(X)$ where objects are bundles over X and morphisms are continuous functions between bundles. Since we are individually assigning a bundle over X to a sheaf we can move a step back and consider the functor

$$\Gamma : \mathbf{Bn}(X) \rightarrow \mathbf{Set}^{\mathcal{O}(X)^{op}}$$

where to each bundle we associate a sheaf and to each morphism between bundles we associate a natural transformation between sheaves.

(Pre)Sheaves to Bundles Let

$$P : \mathcal{O}(X)^{op} \rightarrow \mathbf{Set}$$

be a presheaf. In order to transition from a presheaf over X to a bundle over X we need to do three things:

1. Generate a collection of sets that are indexed by $x \in X$.
2. Take a union over all sets indexed by x and apply some kind of topology.
3. Take this topology and define a map from it to X giving us a bundle over X .

(1) To achieve step 1 we must talk about the notion of a germ of a section over x . Given two open subsets $U, V \subseteq X$ consider $x \in W \subseteq U \cap V$. Now let $s \in P(U)$ and $t \in P(V)$, then if $s|_W = t|_W \in P(W)$ we can say that both of these sections have the same germ at x . This notion actually comes to form equivalence classes of germs such that for any section s there is a germ of s over x denoted $[s]_x$. With the notion of a germ we now have a way of indexing open subsets of X using points of x . Let us collect these germs in a set

$$P_x = \{[s]_x \mid s \in P(U), U \subseteq X, x \in U\}$$

But note that the topology imposed on X gives us open neighborhoods of x . This means that we can take smaller and smaller open neighborhoods without landing on a definite "smallest" set that x describes. Then we want to instead define P_x as the colimit over all open subsets of U containing x .

(2 & 3) Now that we have a way to determine a set indexed by points $x \in X$ we can define

$$\Lambda_P = \bigsqcup_{x \in X} P_x$$

to collect all of these germ sets into one place. We can then define a projection $\pi : \Lambda_p \rightarrow X$. To assign Λ_p a topology we can construct a basis using the equivalence classes of germs of sections. With this topology π is now a continuous function giving us a bundle over X . Note that if we have a natural transformation $h : P \Rightarrow Q$ the disjoint union of function $h_x : P_x \rightarrow Q_x$ induces a map from $\Lambda_P \rightarrow \Lambda_Q$. With this information we can define a functor

$$\Lambda : \mathbf{Set}^{\mathcal{O}(X)^{op}} \rightarrow \mathbf{Bn}(X)$$

where $P \mapsto \Lambda_p$ assigning to each sheaf over X a bundle over X .

0.5 Assigned in Meetings

1. Define the functors $\text{Hom}(-, A)$ and $\text{Hom}(A, -)$ from $\mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$ where $\mathbf{Mod}_R$ represents the category of left R -modules. Show that the functors are well defined.

Proof. **Hom** $(A, -)$ Let R be the unitary ring and A be a left R -module. Define the covariant functor

$$\mathbf{Hom}(A, -) : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$$

where

- Define $\mathbf{Hom}(A, -)(X) = \text{Hom}(A, X)$ for all $X \in \text{Obj}(\mathbf{Mod}_R)$ where $\text{Hom}(A, X)$ is the set of all R -homomorphisms from A to X .
- Let $f \in \text{Mor}(\mathbf{Mod}_R)$ where $f : X \rightarrow Y$. Define $\mathbf{Hom}(A, -)(f) = f_*$. Where $f_* : \text{Hom}(A, X) \rightarrow \text{Hom}(Y, A)$ is given by $f_*(\varphi) = f \circ \varphi$ for all $\varphi \in \text{Hom}_R(A, X)$.

Well Defined: Let us show that $\text{Hom}(A, X)$ is a left R -module. Observe that $\text{Hom}(A, X)$ is an abelian group under pointwise addition. Additionally, if $r \in R$ and $f \in \text{Hom}(A, X)$, we can define scalar multiplication rf as $(rf)(x) = rf(x)$. Let $r, s \in R$ and $f, g \in G$

(i)

$$[r(f + g)](x) = r(f(x) + g(x)) \quad (1)$$

$$= rf(x) + rg(x) \quad (2)$$

$$= (f + g)(rx) \quad (3)$$

(ii)

$$(r + s)f(x) = f((r + s)x) \quad (4)$$

$$= f(rx + sx) \quad (5)$$

$$= rf(x) + sf(x) \quad (6)$$

(iii)

$$(rs)f(x) = f(rsx) \quad (7)$$

$$= r(s(f(x))) \quad (8)$$

(iv)

$$1 \cdot f(x) = f(1 \cdot x) = f(x) \quad (9)$$

Since hom-sets are modules then any then any morphism between them are by definition in $\text{Mor}(\mathbf{Mod}_R)$.

Functor Axioms

Identity: Let $\text{id}_R : R \rightarrow R$, then by the definition of $\mathbf{Hom}_R(A, -)(\text{id}_R) : \text{Hom}(A, R) \rightarrow \text{Hom}(A, R) = \text{id}_{\text{Hom}_R(A, R)}$.

Composition: $\mathbf{Hom}(A, f)(\phi) = f \circ \phi$ for any $\phi \in \text{Hom}_R(A, X)$ then

$$[\mathbf{Hom}(A, f) \circ \mathbf{Hom}(A, g)](\phi) = \mathbf{Hom}(A, g)(f \circ \phi) \quad (10)$$

$$= g \circ (f \circ \phi) \quad (11)$$

$$= (g \circ f) \circ \phi \quad (12)$$

$$= \mathbf{Hom}(A, f \circ g)(\phi) \quad (13)$$

□

$\mathbf{Hom}(-, A)$ Now define the following contravariant functor

$$\mathbf{Hom}(-, A) : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$$

where

- Define $\mathbf{Hom}(-, A)(X) = \text{Hom}(X, A)$ for all $X \in \text{Obj}(\mathbf{Mod}_R)$ where $\text{Hom}(X, A)$ is the set of all R-homomorphisms from X to A .
- Let $f \in \text{Mor}(\mathbf{Mod}_R)$ where $f : X \rightarrow Y$. Where $f^* : \text{Hom}(X, A) \rightarrow \text{Hom}(Y, A)$ is given by $f^*(\phi) = \phi \circ f$ for all $\phi \in \text{Hom}_R(X, A)$.

Well Defined: Because it was show that hom-sets are R-modules and morphisms between hom-sets are R-homomorphisms it suffices that changing the direction will preserve these conditions.

Functor Axioms

Identity: Let $\text{id}_R : R \rightarrow R$, then by definition $\mathbf{Hom}_R(-, A)(\text{id}_R) : \text{Hom}(R, A) \rightarrow \text{Hom}(R, A) = \text{id}_{\text{Hom}_R(R, A)}$.

Composition: $\mathbf{Hom}(-, A)(f \circ \phi) = \mathbf{Hom}(-, A)(f) \circ \mathbf{Hom}(-, A)(\phi)$ for any $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ in $\mathbf{Mor}(\mathbf{Mod}_R)$. Then given any ring homomorphism ϕ

$$[\mathbf{Hom}(f, A) \circ \mathbf{Hom}(g, A)](\phi) = \mathbf{Hom}(g, A)(\phi \circ f) \quad (14)$$

$$= (\phi \circ f) \circ g \quad (15)$$

$$= \phi \circ (f \circ g) \quad (16)$$

$$= \mathbf{Hom}(f \circ g, A)(\phi) \quad (17)$$

Dual Space Let V be a finite dimensional vector space of the field $\mathbb{K}$. Then we can define the dual vector space $V^* = \text{hom}(V, \mathbb{K})$. Without any fancy functors we can construct a set of dual basis of linear functionals from the basis of V that necessarily implies that $V \cong V^*$. In much the same way we can extend this isomorphism to show that $V^* \cong V^{**}$. Because isomorphisms are transitive it follows naturally that $V \cong V^{**}$.

Dual Functor Consider that we have two vector spaces V, W and a transformation $T : V \rightarrow W$. There exists an isomorphism between $f : W \rightarrow W^*$

$$\begin{array}{ccc} V & & \\ \downarrow T & \searrow f \circ T & \\ W & \xrightarrow{f} & \text{Hom}(W, \mathbb{K}) \end{array}$$

By precomposing $f \circ T$ we can define a function $T^* = f \circ T$. Notice that this precomposition returns a function from $V \rightarrow \mathbb{K}$ implying that we get a function in V^* from our composition. Taking a broader view of our diagram

$$\begin{array}{ccc} V & & \text{Hom}(V, \mathbb{K}) \\ \downarrow T & \rightsquigarrow T^* & \uparrow \\ W & & \text{Hom}(W, \mathbb{K}) \end{array}$$

Taking this a step further we can consider the double dual functor. This functor, defined below, takes a dual space to its corresponding double dual space in a similar manner to above.

$$(-)^* : \mathbf{Vect}_{\mathbb{K}}^{\text{Op}} \rightarrow \mathbf{Vect}_{\mathbb{K}}$$

where for any vector space V , $(-)^*(V) \mapsto \text{Hom}(V, \mathbb{K})$ and for any transformation $T : V \rightarrow W$ then $(-)^*(T) \mapsto T^* : \text{Hom}(W, \mathbb{K}) \rightarrow \text{Hom}(V, \mathbb{K})$. In the same way we can continue to define the double dual functor. Given a transformation $T : V \rightarrow W$ and there exists an isomorphism $f : W \rightarrow W^{**}$. Then we can build a new functor, note that this functor is covariant while the dual functor is contravariant, $(-)^{**} : \mathbf{Vect}_{\mathbb{K}} \rightarrow \mathbf{Vect}_{\mathbb{K}}$.

Expanding our diagram $(-)^{**}$

$$\begin{array}{ccccc}
 & & \xrightarrow{(-)^{**}} & & \\
 V & \xrightarrow{\quad \text{Hom}(V, \mathbb{K}) \quad} & & \text{Hom}(\text{Hom}(V, \mathbb{K}), \mathbb{K}) & \\
 \downarrow T & \xrightarrow{\quad (-)^* \quad} & T^* & \downarrow T^{**} & \\
 W & \xrightarrow{\quad \text{Hom}(W, \mathbb{K}) \quad} & & \text{Hom}(\text{Hom}(W, \mathbb{K}), \mathbb{K}) &
 \end{array}$$

Natural Transformation Define the identity functor

$$\text{id} : \mathbf{Vect}_{\mathbb{K}} \rightarrow \mathbf{Vect}_{\mathbb{K}}$$

where $\text{id}(X) = X$ for all $X \in \text{Obj}(\mathbf{Vect}_{\mathbb{K}})$ and $\text{id}(f) = f$ for all $f \in \text{Mor}(\mathbf{Vect}_{\mathbb{K}})$. Let V be a finite dimensional vector space over the field $\mathbb{K}$. For any $v \in V$ there exists a linear functional $f : V \rightarrow \mathbb{K}$ where $f(v)$ is well defined:

$$ev_v : V^* \rightarrow \mathbb{K}, ev_v(v) = f$$

In a similar manner for each $v \in V$ can be assigned to it's corresponding evaluation function. More concretely, define

$$\text{eval}_V : V \rightarrow V^{**}, \text{eval}_V(v) = ev_v$$

I claim that $\text{eval} : \text{id} \Rightarrow (-)^{**}$. To do this I will show that the following diagram commutes

$$\begin{array}{ccc}
 \text{id}(V) & \xrightarrow{\text{eval}_V} & V^{**} \\
 \downarrow \text{id}(f) & & \downarrow f^{**} \\
 \text{id}(W) & \xrightarrow{\text{eval}_W} & W^{**}
 \end{array}$$

For all $v \in V$ where

(a)

$$(\text{id} \circ \text{eval}_W)(v) = \text{eval}_W(\text{id}(w)) \quad (18)$$

$$= \text{eval}_W(w) \quad (19)$$

$$= ev_W \in W^{**} \quad (20)$$

(b)

$$(\text{eval}_V \circ f^{**})(v) = \text{id}(\text{eval}_V(v)) \quad (21)$$

$$= f^{**}(ev_v) \in W^{**} \quad (22)$$

2. Also carefully work through the example of the determinant as a natural transformation. Let GL_n be a functor from the category of commutative rings to the category of groups, where $GL_n(R)$ is the group of invertible $n \times n$ matrices over a ring R . Let $*$ be the functor that maps a ring to its group of units (invertible elements). The determinant map $\det : GL_n(R) \rightarrow R^*$ is a natural transformation between these two functors.

Proof. Let R be a commutative ring and $R^\times$ be its group of units. Define the covariant functor

$$(-)^\times : \mathbf{CRing} \rightarrow \mathbf{Group}$$

where

- Define $(-)^\times(R) = R^\times$ for all $R \in \text{Obj}(\mathbf{CRing})$
- Let $f : R \rightarrow R_0$ be a ring homomorphism. Define $(-)^\times(f) = f|_{R^\times}$ where $f|_{R^\times} : R^\times \rightarrow R_0^\times$.

Well Defined : By definition $R^\times$ contains the invertible elements of R that the underlying group properties of a ring are preserved. Because f is restricted to map to and from a unit group it is guaranteed to be a group homomorphism landing in $\text{Mor}(\mathbf{Group})$.

Functor Axioms Identity: Given the ring homomorphism $\text{id}_R : R \rightarrow R$ then $(-)^\times(\text{id}_R) = \text{id}_R|_{R^\times} = \text{id}_{R^\times}$

Composition: Given the following commutative ring homomorphisms $f : R_0 \rightarrow R_1$ and $g : R_1 \rightarrow R_2$ then for any $\phi \in R_0$

$$(-)^\times(f \circ g)(\phi) = (f \circ g)|_{R_0^\times}(\phi) \quad (23)$$

$$= f|_{R_0^\times} \circ g|_{R_1^\times}(\phi) \quad (24)$$

$$= [(-)^\times(f) \circ (-)^\times(g)](\phi) \quad (25)$$

Now define a new covariant functor

$$\mathbf{GL}_n(\mathbf{R}) : \mathbf{CRing} \rightarrow \mathbf{CRing}$$

where

- Define $\mathbf{GL}_n(R) = GL_n(R)$ where $GL_n(R)$ is the group of invertible $n \times n$ matrices whose elements come from R .
- Let $f : R \rightarrow R'$ be a ring homomorphism between commutative rings R, R' . Then define $\mathbf{GL}_n(f) : GL_n(R) \rightarrow GL_n(R')$.

□

$\mathbf{GL}_n$ and $*$ To start understanding how the determinant behaves in terms of a natural transformation I will first break down how the GL_N and $*$ functors are defined. Let $\mathbf{CRing}$ be the category of commutative rings. Let

$$GL_n : \mathbf{CRing} \rightarrow \mathbf{Group}$$

where for any commutative ring R we define $GL_n(R) \mapsto$ the group of $n \times n$ invertible matrices where elements are from R . To define a functor

between the groups there must some way to transition from a commutative ring to a set of $n \times n$ matrices. We can define a function

$$\theta : R \rightarrow GL_n(R)$$

where $\theta(r) \mapsto rI_n$ where I_n is the $n \times n$ identity matrix. Given that for any ring there exists a unit group $R^\times$. With this we can restrict our map to form a group homomorphism

$$\theta|_{R^\times} : R^\times \rightarrow GL_n(R)$$

where $\theta|_{R^\times}(r) \mapsto rI_n$. There will also exist a group homomorphism $T : GL_n(R) \rightarrow GL_n(R_0)$. With this we can build our diagram

$$\begin{array}{ccc} R & \xrightarrow{\theta|_{R^\times}} & GL_n(R) \\ & \searrow \theta|_{R^\times} \circ T & \downarrow T \\ & & GL_n(R_0) \end{array}$$

which tells us that $GL_n(f) \mapsto f \circ T$. Our other functor

$$* : \mathbf{CRing} \rightarrow \mathbf{Group}$$

mapping a commutative ring to its units where $*(A) \mapsto A^*$ and for any ring homomorphism $\phi : R \rightarrow S$ the functor maps $*(\phi) \mapsto \phi|_{R^*} : R^* \rightarrow S^*$. I think the next question to intuitively ask is how these functors are related. They clearly have the same target and source categories and the same direction arrows. Lets think about our $GL_n(R)$ set. We know that for any $M \in GL_n(R)$ there exists an inverse M^{-1} such that $MM^{-1} = I$. Then because both matrices are invertible we can take the determinants

$$\det(MM^{-1}) = \det(M) \det(M^{-1}) = 1$$

Because the input for our matrices come from a ring the determinant is mapping an invertible matrix to a ring element. In particular the determinant is mapping a matrix to a unit of a ring. We can say that for any ring R we have a function $\det_R : GL_n(R) \rightarrow R^*$. Where for any $M \in GL_n(R)$ our function $\det_R(M) = a \in R^*$. This collection of functions will serve to define the natural transformation $\det : GL_n \Rightarrow *$. To determine whether this is a well defined natural transformation let $f : R \rightarrow S$ be a ring homomorphism then

(a)

$$\begin{aligned} GL_n(f) \circ \det_Y &= (GL_n(X) \rightarrow GL_n(Y)) \circ (GL(Y) \rightarrow Y^*) \\ &= GL_n(X) \rightarrow Y^* \end{aligned}$$

(b)

$$\begin{aligned}\det_X \circ f^* &= (GL_n(X) \rightarrow X^*) \circ (X^* \rightarrow Y^*) \\ &= GL_n(X) \rightarrow Y^*\end{aligned}$$

implying that our natural transformation is well defined.

0.6 Textbook Exercises

1. **Skyscraper Construction** Let X be a topological space and $x \in X$. Let S be a set and $i_p : p \rightarrow X$ an inclusion. Define

$$i_{p,*}F(U) = \begin{cases} S & \text{if } p \in U, \\ \{e\} & \text{if } p \notin U \end{cases}$$

Proof. Consider the possible restriction functions. For any two subset $U \subseteq V$ define the restriction $\rho|_V^U : F(U) \rightarrow F(V)$ such that for all $s \in F(U)$, $\rho|_V^U(s) = s|_V$, then we have three cases.

- (a) If $p \in U \cap V$ then $\rho|_V^U : S \rightarrow S$ is the identity. Then for any U_i, U_j if $p \in U_i \cap U_j$ then the restrictions to U_i and U_j will agree.
- (b) If $p \in U$ but $p \notin V$ then $\rho|_V^U : S \rightarrow 0$. Then for any U_i, U_j with the above conditions it follows that $p \notin U_i \cap U_j$. The the sections restricted to both U_i and U_j will both map to zero.
- (c) If $p \notin U, V$ then $\rho|_V^U : 0 \rightarrow 0$. In this case it is immediate to see that the restrictions of any sections under these conditions will map to zero.

Gluing is satisfied, it remains to show that these sections are unique. Let $t, s \in F(U)$ by definition there exists a cover $\{U_i\}_{i=1}^n$ such that $U = \bigcup_{i=1}^n U_i$. If $p \in U_i$ then $s|_{U_i} = S = t|_{U_i}$ otherwise if $p \notin U_i$ then $s|_{U_i} = \{e\} = t|_{U_i}$. Therefore $s = t$. $\square$

2. (CITE The Rising Sea Exercise 2.2H) Suppose $\pi : X \rightarrow Y$ is a continuous map, and F is a presheaf on X . Define π_*F on Y such that $\pi_*F(V) = F(\pi^{-1}(V))$, where V is a subset of Y . (a) Show that π_*F is a presheaf on Y and (b) is a sheaf if F is a sheaf.

Proof. (a) Show that π_*F is a pre-sheaf. Define $\pi_*F : \mathcal{O}(Y)^{op} \rightarrow \mathbf{Set}$ where for any open subset $V \subseteq Y$, $\pi_*F(V) \mapsto F(\pi^{-1}(V))$ for an inclusion of open subsets $U \subseteq V$ the functor $\pi_*F(i) \mapsto \rho|_{\pi^{-1}(V)}^{\pi^{-1}(U)}$. Given that F is a pre-sheaf π_*F is a well-defined functor. Given $\text{id}_U : U \hookrightarrow U$ it follows

$$\pi_*F(\text{id}_U) = \rho|_{\pi^{-1}(U)}^{\pi^{-1}(U)} = \text{id}_{\pi_*F(U)}$$

Let U, V, Z be open subsets of X with maps $\alpha : U \hookrightarrow V$, $\beta : V \hookrightarrow Z$. Let $s \in \pi_* F(Z)$. Then,

$$[\pi_* F(\beta) \circ \pi_* F(\alpha)](z) = \pi_* F(\alpha)(z)|_{\pi^{-1}(Z)}^{\pi^{-1}(V)} \quad (26)$$

$$= z|_{\pi^{-1}(Z)}^{\pi^{-1}(V)}|_{\pi^{-1}(V)}^{\pi^{-1}(U)} \quad (27)$$

$$= z|_{\pi^{-1}(Z)}^{\pi^{-1}(U)} = \pi_* F(\beta \circ \alpha)(z) \quad (28)$$

Then $\pi_* F$ is a pre-sheaf.

- (b) Suppose that F is a sheaf. For any open set U of Y there exists a covering set $\{V_i\}_{i \in I}$ indexed by a family I . Since unions are preserved under inverses of continuous functions for $\pi^{-1}(U)$ there exists a covering set $\{\pi^{-1}(V_i)\}_{i \in I}$. Since F is a sheaf we have that for any $s_i \in F(\pi^{-1}(V_i))$ and for any i, j the restrictions

$$\rho|_{\pi^{-1}(V_i)}^{\pi^{-1}(V_i \cap V_j)}(s) = \rho|_{\pi^{-1}(V_j)}^{\pi^{-1}(V_i \cap V_j)}(s)$$

by definition implies that there exists an $s \in F(\pi^{-1}(U))$ such that $\rho|_{\pi^{-1}(V_i)}^U(s) = s_i$

□